

Group Topic: Student Dropout Prediction & At-Risk Monitoring System

Lab No.: 2

Student Name: Christel Mae Capa

Individual Contribution: Dataset Research & Justification

Git Repo / Colab Link:

<https://github.com/dgbb-prog/student-dropout-bigdata-datacleaning>

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Objectives

The primary objective of this week's task is to identify, evaluate, and justify a suitable dataset for the group's application development project. Specifically, the goal is to:

- Source a publicly available, ethically compliant dataset with a minimum of 500,000 rows.
- Verify that the dataset is relevant to the application the group intends to build a Student Dropout Prediction and At-Risk Monitoring System.
- Document the rationale behind the dataset selection, including its origin, licensing, structure, and suitability for data analysis and machine learning.
- Ensure the dataset meets the data ethics requirements of the course, including proper attribution and open licensing.

Tasks Completed

The following tasks were accomplished for this week's contribution:

1. Researched multiple publicly available educational datasets suitable for a student dropout prediction application. Platforms explored include Kaggle, UCI Machine Learning Repository, Google Dataset Search, and the Open University Learning Analytics Dataset (OULAD) portal.
2. Evaluated each candidate dataset based on four criteria: (a) row count of at least 500,000, (b) ethical and legal availability, (c) relevance to student behavior and academic outcomes, and (d) completeness and variety of features.
3. Selected the OULAD (Open University Learning Analytics Dataset) from <https://analyse.kmi.open.ac.uk/open-dataset> as the final dataset. The dataset is released under the Creative Commons Attribution 4.0 (CC-BY 4.0) license, making it fully open and ethical for academic use.
4. Documented the structure of the OULAD dataset, which consists of 7 interrelated CSV files: studentInfo.csv, studentVle.csv, studentAssessment.csv, studentRegistration.csv, assessments.csv, vle.csv, and courses.csv. The combined total across all files exceeds 500,000 rows, satisfying the minimum data requirement.
5. Communicated the dataset choice and its justification to the group, ensuring alignment with the application's goals before the group proceeded to data cleaning and processing.

Tasks to be Completed

Results and Analysis

The dataset research phase yielded the following results:

The OULAD dataset was identified as the most suitable option among the candidates reviewed. It contains rich, multi-dimensional data capturing student demographics, online learning behavior (VLE interactions), assessment scores, and final academic outcomes across multiple course presentations.

Key dataset statistics confirmed for the group:

- studentInfo.csv — demographic and enrollment data per student
- studentVle.csv — virtual learning environment interaction logs (largest file, primary source of behavioral data)
- studentAssessment.csv — individual assessment submission records and scores
- studentRegistration.csv — enrollment and withdrawal dates
- assessments.csv, vle.csv, courses.csv — supporting reference tables

Combined total rows across all files: well above 500,000 rows. Requirement is satisfied.

The dataset's final_result column (Pass, Fail, Distinction, Withdrawn) serves as the primary target variable for the group's dropout prediction model, with 'Withdrawn' mapped to a high-risk label. This directly supports the application's goal of identifying at-risk students early.

Challenges and Solutions

Challenge: Finding a dataset that met all four criteria simultaneously (500k+ rows, ethical license, educational domain, rich features) was initially difficult. Several Kaggle datasets were either too small or lacked proper licensing information.

Solution: Expanded the search beyond Kaggle to institutional and academic data repositories. The OULAD portal, maintained by the Knowledge Media Institute of The Open University, provided a fully documented, CC-BY 4.0 licensed dataset that satisfied all requirements.

Challenge: Verifying that the combined row count across multiple files met the 500,000 minimum without running the full code.

Solution: Reviewed the official OULAD documentation and dataset description page, which provided the row counts and file sizes. The studentVle.csv file alone contains over 10 million rows, far exceeding the requirement.

Conclusion

The dataset research and justification task was completed successfully. The OULAD dataset was selected as the group's primary data source due to its size (500,000+ rows across 7 files), ethical licensing (CC-BY 4.0), educational

relevance, and richness of features covering student demographics, learning behavior, and academic outcomes.

This week's task reinforced the importance of thorough dataset evaluation before proceeding to data processing. Selecting the right dataset at the outset ensures that the group's application a Student Dropout Prediction and At-Risk Monitoring System is built on a solid, reliable, and ethically sound data foundation. The dataset is now ready for the group's cleaning, processing, and visualization pipeline.